

Day 10 – Advanced IAM Security & Secure Cloud Architecture Project

Project Objective

Build a secure AWS architecture using EC2, IAM Roles, S3, CloudWatch, MFA, and Security Groups.

Technologies Used

AWS EC2, IAM, S3, CloudWatch, CloudTrail, CloudFront, Load Balancer.

Architecture Overview

Secure user traffic flows through CloudFront and Load Balancer into EC2 with IAM-based access.

IAM Roles

EC2 securely accesses S3 without hardcoded access keys.

Security Features

MFA, Security Groups, Least Privilege Access, Monitoring, Logging.

CloudWatch

Tracks metrics, logs, alarms, and suspicious activities.

CloudTrail

Records API calls and user activities for auditing.

Hands-On Tasks

Create EC2, Configure IAM Role, Attach Policies, Enable MFA, Configure CloudWatch Alarm.

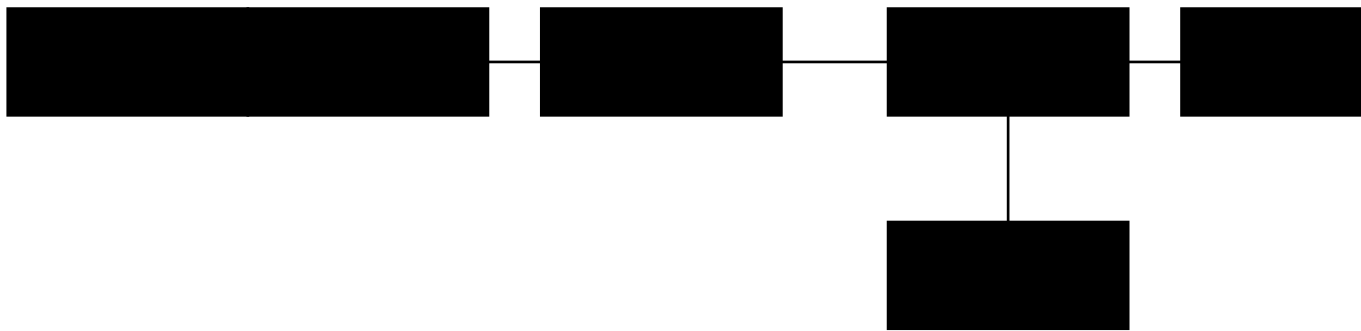
Use Cases

Banking apps, SaaS platforms, AI pipelines, enterprise systems.

Learning Outcomes

Learned enterprise-grade AWS security and monitoring practices.

Architecture Diagram



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